

DATA MINING PROJECT

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Segmentation for Personalized Marketing Amazing International Airlines Inc.

Individual Project

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TABLE OF CONTENTS

1. Introduction	1
2. DATA PREPARATION.....	1
2.1. Missing Values, Redundancy, Data Type Issues, and Ambiguities	1
2.2. Duplicates	2
2.3. Outlier analysis	2
3. PREPROCESSING	3
3.1. Data cleaning	3
3.2. Feature Engineering and Feature Enrichment	3
3.3. Feature scaling.....	3
3.4. DBSCAN multivariate outliers.....	4
3.5. Feature selection	4
3.6. Clustering methodology and model selection.....	5
4. Mapping Consolidated Clusters to Strategic Marketing Priorities.....	6
5. Cluster visualization	9
6. FEATURE IMPORTANCE ASSESSMENT.....	9
7. Conclusion	10
Bibliographical References	11
Appendix.....	11
Appendix A - AI Usage Note	11
Appendix B - Contribution Statement.....	11
Appendix C - RESPONSIBILITY Statement.....	11

1. INTRODUCTION

This project focuses on customer segmentation for Amazing International Airlines Inc. (AIAI) using historical customer and flight data. The main objective is to identify distinct customer groups based on their value, engagement, and travel behavior, and to translate these patterns into actionable business insights.

After an initial exploratory analysis, clustering techniques were applied to better understand how customers differ in terms of lifetime value, flight activity, travel distance, companion behavior, and loyalty engagement. The analysis combines multiple behavioral dimensions to capture meaningful differences across customers. Clustering was performed using K-Means, Hierarchical Clustering, and Self-Organizing Maps (SOMs), while DBSCAN was used specifically for multivariate outlier detection at the appropriate stage of the pipeline (after feature preparation). The final goal is to support more targeted strategies by focusing on segments with the highest strategic impact.

2. DATA PREPARATION

2.1. Missing Values, Redundancy, Data Type Issues, and Ambiguities

Data preparation started with resolving inconsistencies, ambiguities, and structural issues that could affect distance-based clustering. In the **cancellation date** field, the value **"29/02/2019"** appeared in the data even though 2019 is not a leap year. This value was corrected to **"28/02/2019"**, after which the entire cancellation date column was converted to a proper datetime format using day-first parsing. This ensured that all date-related operations relied on valid calendar values.

Several variables were removed due to redundancy or lack of analytical value. An **"Unnamed"** column duplicated the index and was therefore discarded. Customer identification fields such as **first name**, **last name**, and **full name** were also removed, as they do not describe customer behavior or value and would only introduce noise into the clustering process. In addition, the **Country** variable was excluded because it contained a single constant value (**Canada**) for all customers, making it non-informative for segmentation.

Missing values were handled selectively, depending on the importance of each variable. **Customer Lifetime Value (CLV)** is a central feature in value-based segmentation, and observations with missing CLV cannot be interpreted reliably. In total, **20 records** with missing CLV were removed to avoid distorting cluster structure. These records were later confirmed to coincide with additional data quality issues, reinforcing the decision to exclude them.

The **Income** variable was removed entirely due to clear reliability problems. Its distribution varied unrealistically across education levels: for example, Income values were **zero for all customers with College education**, while values for **Master** and **Doctor** levels were compressed into a narrow range, unlike the more plausible distribution observed for **Bachelor**. This pattern suggests encoding or clipping issues rather than genuine economic differences. Since Income could not be used consistently across customers, it was excluded from the analysis to prevent bias in distance calculations.

2.2. Duplicates

Duplicate records were handled with careful consideration of the different granularities present in the data. Since customer information and flight activity are stored at different levels, duplicates were identified and treated separately for each table.

At the customer level, duplicates were defined by repeated values of **Loyalty#**, which uniquely identifies each customer. Removing these duplicates reduced the customer table from **16,901 to 16,574** observations. This step ensured that each customer appeared only once in the modeling dataset.

At the flight level, duplicates were defined by repeated combinations of **Loyalty#**, **Year**, and **Month**, which violate the expected monthly aggregation structure. Removing duplicated customers at the customer level propagated naturally to the flight table, reducing it from **608,436 to 596,664** records.

A consistency check confirmed that the set of duplicated **Loyalty#** values was identical in both tables. Each duplicated customer appeared exactly **36 times** in the flight table, corresponding to the **36 months between January 2019 and December 2021**. This confirms that duplicates originated from repeated customer records rather than genuine repeated flight activity. Removing them prevented artificial inflation of cumulative metrics such as total flights, total distance, and redeemed points.

2.3. Outlier analysis

Outlier analysis was conducted using both univariate and bivariate perspectives, with the goal of identifying problematic observations while preserving meaningful extreme behavior that is relevant for customer segmentation.

Univariate analysis was performed using distributional inspection and boxplots for key numeric variables, including **Customer Lifetime Value**, **number of flights**, **total distance flown**, and **dollar value of points redeemed**. Extreme values were not automatically treated as errors, as high values in these variables often correspond to highly engaged or high-value customers. Therefore, univariate outliers were primarily used for diagnostic purposes, to understand the range and skewness of the data, rather than as a strict removal criterion.

Bivariate analysis focused on identifying inconsistencies arising from the relationship between variables. A specific group of observations exhibited simultaneous issues across multiple dimensions: the same **city** appeared with **multiple latitude–longitude pairs**, indicating ambiguous or incorrect spatial information. Notably, these observations were also the **only records with missing Customer Lifetime Value and missing Income**. In total, **20 records** showed this combined pattern of spatial inconsistency and missing core variables. Given the presence of multiple data quality issues and the limited size of this group, these records were removed from the dataset.

Density-based methods such as **DBSCAN** were not applied at this stage for outlier detection. DBSCAN relies on distance computations and therefore requires all metric features to be properly scaled. At this point in the pipeline, feature scaling had not been applied, making distance-based outlier detection unreliable. For this reason, DBSCAN-based multivariate outlier detection was deferred to later stages of the analysis, after feature engineering and normalization were completed.

3. PREPROCESSING

Preprocessing aimed to ensure that clustering results reflect real customer behavior and value patterns rather than artifacts of data quality, scale, or aggregation. This stage includes data cleaning, feature engineering and enrichment, feature scaling, and multivariate outlier detection using DBSCAN.

3.1. Data cleaning

Data cleaning consolidated and finalized the consistency checks performed earlier. Customer-level and flight-level datasets were aligned to ensure that all subsequent feature construction relied on coherent and comparable inputs. After removing duplicates and resolving inconsistencies, all remaining records were validated against expected data ranges and logical constraints.

Special attention was paid to variables later used in aggregation, such as flight counts, distance, redeemed points, and tenure-related fields. These variables were verified to contain no negative values or logically impossible combinations. At this stage, the dataset contained a single, clean representation of each customer and their associated flight history, suitable for feature construction.

3.2. Feature Engineering and Feature Enrichment

Feature engineering focused on transforming raw flight activity into meaningful customer-level descriptors that capture engagement intensity, value, and travel behavior. Since clustering is performed at the customer level, flight-level variables were aggregated accordingly.

The total number of flights, total distance flown, and total dollar value of points redeemed were computed per customer to represent cumulative engagement and monetary interaction with the airline. It worth noting that **number of flights with companions** reflects group travel behavior and potential revenue leverage beyond individual customers.

Customer tenure was expressed in **days**, providing a continuous measure of relationship duration rather than a coarse categorical proxy. This allowed tenure to interact naturally with cumulative features such as total flights and lifetime value.

Loyalty-related variables were preserved to support interpretation and post-clustering analysis. Although loyalty attributes were not used directly for clustering, they provide essential context for understanding differences between segments and for defining strategic actions later in the analysis.

Feature enrichment was deliberately conservative: only variables with clear behavioral meaning and sufficient data quality were retained. Variables excluded earlier, such as Income, were not reintroduced at this stage to avoid contaminating distance-based models with unreliable signals.

3.3. Feature scaling

Before applying clustering and distance-based methods, all metric features were scaled using **Min–Max normalization**. This step was necessary because key variables, such as Customer Lifetime Value, number of flights, and total distance, are expressed in very different units and ranges.

Without scaling, high-magnitude variables would dominate distance calculations and bias clustering outcomes. Min–Max scaling preserves relative differences between customers while ensuring that each feature contributes proportionally to similarity computations. Scaling was applied only after feature engineering was completed, so that derived variables were treated consistently with original ones.

3.4. DBSCAN multivariate outliers

After feature engineering and scaling, **DBSCAN** was applied for **multivariate outlier detection**, not for clustering. Importantly, DBSCAN was run **only on scaled metric features**, excluding categorical variables entirely. This ensured that distance calculations were meaningful and not distorted by arbitrary encodings.

DBSCAN was used to identify customers whose overall behavior deviated strongly from the main population across multiple dimensions simultaneously. Unlike univariate inspection, this approach captures rare combinations of otherwise valid values that may indicate abnormal or unreliable observations.

Customers identified as noise by DBSCAN were removed from the dataset at this stage. This step reduced the influence of extreme multivariate profiles that could distort centroid-based clustering methods, while preserving meaningful high-value or high-engagement customers that remain consistent across dimensions.

3.5. Feature selection

Feature selection was performed after preprocessing and feature construction to reduce redundancy and ensure that clustering is driven by complementary behavioral signals rather than overlapping information. Since distance-based clustering methods are sensitive to correlated inputs, careful attention was paid to multicollinearity among metric features.

Before correlation analysis, several numeric variables exhibiting strong right skew—such as **Customer Lifetime Value**, **total distance flown**, and **number of flights**—were transformed using a logarithmic transformation. This step reduced extreme skewness and stabilized variance, improving comparability across customers. However, even after log transformation and Min–Max scaling, some features remained moderately skewed and clearly non-Gaussian.

For this reason, **Spearman rank correlation** was used instead of Pearson correlation. Spearman correlation is more appropriate in this context, as it captures monotonic relationships and is less sensitive to residual skewness and extreme values. This choice ensured that feature dependence was assessed reliably without relying on the assumption of linear relationships or normality.

Correlation matrices were inspected to identify highly correlated feature pairs. When strong correlations were detected, features were evaluated based on interpretability and strategic relevance rather than removed automatically. In particular, cumulative engagement variables such as **number of flights**, **total distance flown**, and **Customer Lifetime Value** were expected to be correlated by construction, yet they were not fully redundant. Each captures a different dimension of customer behavior—frequency, intensity, and monetary value—and therefore selected features were retained selectively to preserve these distinctions.

Variables that were purely identificational or descriptive were excluded from the clustering feature set. In addition, categorical variables were not included in the distance-based clustering input. Instead, they were retained for **post-clustering profiling and interpretation**, allowing clusters to be described in business terms without introducing distortions through arbitrary numeric encoding.

As a result, the final feature set used for clustering represents a balanced combination of value, engagement, and behavioral intensity, while minimizing redundancy and ensuring robustness to skewed distributions.

3.6. Clustering methodology and model selection

Clustering was conducted through a multi-step approach that combines different analytical views of customer behavior and value. Rather than applying a single clustering model to all features at once, the analysis was structured around **three complementary clustering views**: one **value-based view** and two **behavior-based views**, each capturing a distinct aspect of customer activity. This separation allowed each clustering solution to focus on a coherent feature space and reduced the risk of mixing heterogeneous signals in a single distance computation.

For each view, clustering was performed independently using **K-Means**, **Hierarchical Clustering**, and **Self-Organizing Maps (SOMs)**. K-Means served as a baseline method due to its scalability and interpretability. Multiple values of k were tested, and model performance was evaluated using **inertia** and **silhouette scores**. Inertia was used to assess within-cluster dispersion and to identify elbow points where additional clusters no longer resulted in meaningful error reduction. Silhouette scores were used to evaluate cluster separation and cohesion, ensuring that customers were both close to their assigned cluster and well separated from others.

Hierarchical clustering was applied to further explore the internal structure of each view. Several linkage criteria were tested, including **single linkage**, **complete linkage**, **average linkage**, and **Ward's linkage**. Single linkage was evaluated for chaining effects, while complete and average linkage provided alternative compactness assumptions. Ward's linkage was particularly relevant due to its variance-minimizing property and was closely examined through dendrograms. Hierarchical solutions were compared using inertia and silhouette diagnostics, in parallel with K-Means results.

Self-Organizing Maps were then used to project each feature space onto a lower-dimensional grid while preserving neighborhood relationships. SOMs enabled visual inspection of customer similarity patterns that are not easily observable through centroid-based methods. **Component planes** were used to analyze feature distributions across the map, while **Hits Maps** and the **U-Matrix** helped identify dense regions and natural separations. To obtain explicit cluster labels, both K-Means and Hierarchical Clustering were applied on top of the SOM grid, allowing direct comparison with non-SOM-based solution (K-Means++).

For each view, clusters were **profiled immediately after clustering**, using both numeric summaries and categorical distributions. This intermediate profiling step ensured that clusters were interpretable on their own and provided a clear basis for subsequent integration across views.

After view-specific clustering, clusters were **merged across views** to form the final segmentation. Importantly, the merged clusters showed a **clear and consistent correspondence** with the original clusters they were derived from. No structural contradictions were observed between the initial cluster assignments and the merged results. As a result, merging reduced segmentation complexity while **preserving the quality, separation, and interpretability** of the original solutions.

The final outcome consists of **six distinct customer clusters**. These clusters retain the behavioral and value-based characteristics identified in the initial views, while offering a more compact and actionable segmentation. Final profiling confirmed that the merged clusters remain well differentiated and aligned with the patterns observed throughout the earlier stages of the analysis.

4. MAPPING CONSOLIDATED CLUSTERS TO STRATEGIC MARKETING PRIORITIES

Based on the preceding analysis, we identified six distinct groups of individuals with different characteristics.

Table 1. Consolidated Clusters and Their Key Characteristics

	CLV	Points Redeemed (\$)	Tenure	Flights	Flights w/ Companions	Distance	Active	Aurora	Star	2021 Promo
C0	High	Low	High	Medium-High	Medium	Medium-High	Very High	Low-Medium	Medium	Low
C1	Low	Medium-High	Medium	Low	Low	Low-Medium	High	Low	High	Medium
C2	High	High	High	High	Medium-High	High	Very High	Medium	Medium	Low
C3	Low-Medium	Low	Low-Medium	Very Low	Very Low	Very Low	Medium	Low	Medium-High	Low-Medium
C4	Low-Medium	Low	High	High	Medium	High	Very High	Low	Medium-High	Low
C5	Low-Medium	High	High	High	High	High	Very High	Low-Medium	Medium-High	Low

For each consolidated cluster, a strategic focus is defined, followed by concrete operating-level initiatives outlining how the strategy can be implemented in practice.

Cluster 0 – Long-Term Regulars (high-value, long-tenure customers with moderate cumulative usage). This cluster is characterized by high customer lifetime value and long tenure. Customers show medium to high cumulative flight activity and cumulative distance, reflecting sustained participation over time rather than extreme usage. Activity status is high, indicating strong retention. Points redemption is low, and companion travel is moderate. This cluster is distinguished by value accumulation driven primarily by longevity and retention rather than intensive usage or reward redemption.

Strategic focus: keep long-term, high-value customers engaged and loyal.

- For this cluster, focus on retaining customers whose value is driven primarily by long-term membership rather than intensive usage.
- Strengthen loyalty tier benefits (*e.g., Aurora privileges*) so customers feel recognized and rewarded for staying with AIAA.
- Look for careful upsell opportunities without relying on heavy discounts (*e.g., value-added services such as premium seating or flexible tickets*).
- Support repeat travel with companions to maintain regular travel patterns (*e.g., companion bonuses, shared benefits*).

Cluster 1 – At-Risk Low-Value Customers (*low-value, medium-tenure customers with limited accumulated usage*). This cluster includes customers with low customer lifetime value and low cumulative flight activity, as reflected by low total flights and low cumulative distance. Tenure is moderate, while activity status remains relatively high, suggesting that many customers are still active but have not converted their time in the program into substantial flight activity or value accumulation.

The relatively high level of points redemption compared to usage may indicate a greater sensitivity to rewards and short-term benefits. Companion travel is low, and loyalty status is predominantly Star, indicating low-tier participation. The defining characteristic of this cluster is continued membership with limited accumulated activity and value to date.

Strategic focus: prevent churn by converting continued membership into meaningful engagement while tightly controlling incentive costs.

- Target this cluster with simple activation campaigns aimed at increasing flight frequency (*e.g., milestone prompts, “one-more-flight” messages*).
- Use limited, usage-linked incentives rather than unconditional discounts (*e.g., rewards unlocked only after a completed trip*).
- Emphasize clear and transparent loyalty mechanics (*e.g., simple explanations of how flights translate into points or tier progress*).
- Deprioritize companion-based offers, as companion travel is low and unlikely to drive near-term impact; social travel habits are not yet developed.

Cluster 2 – Elite Power Users (*top-value customers with very high cumulative usage and reward activity*). This cluster represents customers with the highest customer lifetime value, long tenure, and very high cumulative flight activity, reflected in both total flights and cumulative distance. Activity status is very high, indicating strong retention. Points redemption is high, and companion travel is moderate to high. Loyalty status is most concentrated in the Aurora tier ($\approx 32\%$), the highest share among all clusters. This cluster is uniquely defined by the combination of very high accumulated value, very high cumulative usage, and strong reward utilization.

Strategic focus: protect and strengthen relationships with the most valuable customers.

- Prioritize retention of this group (especially Aurora-tier customers), as they contribute the most value to AIAI (*e.g., proactive retention monitoring, priority handling of service issues*).
- Focus on service quality, recognition, and exclusive benefits rather than price promotions (*e.g., priority services, exclusive access or privileges*).
- Reinforce elite loyalty status to maintain strong commitment to the brand (*e.g., clear communication of elite benefits, visible tier recognition*).
- Leverage companion and social travel patterns to support continued engagement (*e.g., shared premium experiences, companion privileges aligned with elite status*).

Cluster 3 – Dormant High-Value (*lapsed or at-risk former high-value customers; customers who generated comparable revenue in the past, as reflected by customer lifetime value, but who now show very low recent activity and a low to moderate activity status*). This indicates a shift from previously high customer value to dormancy or near churn. These customers represent former high-value users who are currently disengaged and should be considered lapsed or at-risk, making them suitable candidates for selective reactivation efforts.

Strategic focus: Attempt reactivation carefully and efficiently.

- Target only customers who previously generated high value (*e.g., former high-tier or high-CLV members*).
- Remind them of past loyalty benefits and status to encourage return (*e.g., references to prior tier privileges or lost benefits*).
- Use incentives selectively and stop efforts if customers do not respond.
- Avoid spending heavily on customers who remain inactive.

Cluster 4 – **Frequent but Under-Monetized** (*high-usage, long-tenure customers with low reward utilization and modest CLV*). This cluster consists of customers with high cumulative flight activity and high cumulative distance, accumulated over long tenure. Activity status is very high, indicating strong retention. However, customer lifetime value is only low to moderate, and points redemption is low, while companion travel is moderate. This cluster is distinguished by sustained high usage over time without proportional value accumulation or reward engagement.

Strategic focus: Convert high usage into higher value.

- Improve value capture from frequent travel (*e.g., encourage higher-value ticket choices or selected paid add-ons*).
- Increase awareness of loyalty rewards and how customers can benefit from them.
- Encourage loyalty tier progression where appropriate (*e.g., highlighting how close customers are to the next tier*).

Cluster 5 – **Social Heavy Travelers (Under-Monetized)** (*very high-usage customers with strong companion and reward activity*) This cluster includes customers with very high cumulative flight activity and very high cumulative distance, accumulated over long tenure. Activity status is very high, indicating strong retention. Points redemption is high, and companion travel is high, clearly separating this cluster from others with similar usage levels. Despite this, customer lifetime value remains low to moderate. The key differentiator of this cluster is intensive accumulated usage combined with frequent companion travel and strong reward redemption, despite only low to moderate CLV.

Strategic focus: Manage high engagement while protecting profitability.

- Treat companion travel as a high-leverage growth opportunity (*e.g., companion bonuses, group booking incentives*).
- Actively use rewards and incentives to encourage social travel behavior (*e.g., extra points for bringing companions, earning points together*).
- Apply clear limits to companion-based offers to control costs (*e.g., tiered group benefits, limits on companion bonuses*).

While the clustering analysis provides a comprehensive view of customer structure, **the following takeaways focus on the segments and behaviors with the highest strategic impact:**

- Focus on customers who frequently travel with companions, as they represent a high-leverage opportunity: each additional companion generates incremental revenue, and shared travel behavior reinforces future flight frequency.
- Actively use reward mechanisms to support companion travel, as these customers tend to redeem rewards more frequently, indicating that incentives are effective in encouraging social travel.
- Address disengagement among previously high-value customers, as declining activity may reflect changing customer expectations, new competitive offerings, or broader market shifts, signaling a need to reassess current loyalty benefits and engagement approaches.

A brief geographic review was also conducted. While customer distribution across regions is broadly consistent, Yukon emerges as a notable outlier. Despite its small customer base, Yukon shows relatively high customer lifetime value per customer; this is notable given recent population growth from approximately **40,000 residents in 2021** (*Wikipedia contributors, 2025*) to **48,261 residents** in the fourth quarter of 2025 (*Statistics Canada, 2025*), this suggests potential for selective business

development, provided that expansion remains proportional to market size and operational constraints.

5. CLUSTER VISUALIZATION

Cluster visualization was used as a validation and interpretation tool to assess how well the final six clusters separate in lower-dimensional space. Regarding merged clusters, since clustering was performed in a higher-dimensional feature space, dimensionality reduction techniques were applied solely for visualization purposes, not for model training or selection.

Principal Component Analysis (PCA) was first applied to project the data into two dimensions. PCA provided a linear projection that captures the largest share of variance, allowing an initial inspection of global structure. While PCA made broad differences between clusters visible, substantial overlap remained, which is expected given the linear nature of the transformation and the complexity of customer behavior patterns.

To better capture non-linear relationships, **t-distributed Stochastic Neighbor Embedding (t-SNE)** was applied directly to the scaled feature set. Compared to PCA, t-SNE produced clearer local separation between clusters and revealed meaningful neighborhood structure, particularly for high-usage and high-value customers. This visualization confirmed that customers assigned to the same cluster tend to form compact groups in non-linear space, while different clusters occupy distinct regions.

t-SNE was also applied on **PCA-reduced data** (t-SNE based on PCA) to test whether prior dimensionality reduction would improve separation or stability. However, this approach did not produce clearer or more interpretable results than applying t-SNE directly to the original scaled features.

Uniform Manifold Approximation and Projection (UMAP) was then used as an alternative non-linear visualization technique. UMAP provided a comparable global structure with t-SNE, while still preserving local relationships. The six consolidated clusters appeared well enough separated, with smooth transitions between neighboring groups, making UMAP particularly useful for interpreting how clusters relate to one another rather than viewing them as isolated segments.

As with t-SNE, UMAP was also applied after PCA reduction. However, **UMAP based on PCA did not improve cluster separation or interpretability** compared to applying UMAP directly to the scaled feature set. The additional PCA step reduced information relevant for visualization, leading to less distinct cluster boundaries.

Overall, **t-SNE and UMAP applied directly to scaled metric features** provided the most informative visual validation of the clustering results. These visualizations confirmed that the six final clusters are well structured, internally coherent, and consistent with the behavioral and value-based differences identified during clustering and profiling.

6. FEATURE IMPORTANCE ASSESSMENT

Feature importance and cluster quality were assessed using two complementary evaluations: **Cluster Separation Power (SS-based R^2)** and **Predictive Power for Cluster Membership (Model-Based R^2)**. These checks were performed after clustering, using the final consolidated cluster labels.

Cluster separability was evaluated through **SS-based R^2** , which measures how much of the overall variance in the feature space is explained by differences between clusters. The merged cluster labels achieved better separability than the initial clustering views and any individual feature, supporting that **cluster merging reduced complexity while retaining (and improving) overall cluster quality**.

To assess which variables most strongly explain cluster membership, **Model-Based R^2** was computed for the six numeric features used to construct the merged clusters. This analysis shows that cluster membership is most strongly explained by **DollarCostPointsRedeemed** and **NumFlights**, followed by **DistanceKM** and **NumFlightsWithCompanions**. **tenure_days** contributes moderately, while **Customer Lifetime Value** has the weakest explanatory power among the six features in this model-based evaluation. Overall, this indicates that the consolidated segmentation is primarily driven by **reward redemption and accumulated flight activity**, with travel intensity and social travel behavior adding further differentiation.

Outliers detected using **DBSCAN on scaled metric features** were excluded during clustering to avoid distorting distance-based solutions. Given their limited number, they were later **assigned to the consolidated clusters** using supervised classification based on the same six numeric features. This ensured that every customer received a single final cluster label without weakening the integrity of the clustering solution.

From an operational perspective, the stronger performance of the **merged cluster labels** confirms that merging was successful: it simplifies segmentation while preserving structure. This allows decision-making to rely on **one consolidated cluster per customer**, rather than combining multiple intermediate labels through rule-based schemes. Rule-based approaches still have value for exploratory and flexible targeting, but the merged-cluster framework offers **lower complexity and a more objective, statistics-driven segmentation** for reporting and scalable use.

7. CONCLUSION

This project applied a structured clustering framework to segment AIAI customers based on value and behavior. By first analyzing value-based and behavior-based cluster solutions separately, and then consolidating them through cluster merging, the analysis achieved a more compact and coherent segmentation.

Quantitative evaluation using cluster separation metrics and predictive power confirmed that the **merged cluster solution outperforms the individual clustering views**, demonstrating stronger separability while reducing complexity. Importantly, the merged clusters preserve the behavioral and value-based patterns identified in the initial solutions, allowing each customer to be assigned to a single, well-defined segment without relying on rule-based combinations.

The final six clusters provide a clear and actionable representation of the customer base. This structure supports objective, non-overlapping segmentation suitable for strategic analysis and operational decision-making, while still allowing flexibility through exploratory or rule-based approaches when needed. Overall, the results show that consolidating clustering perspectives leads to higher-quality insights and a more practical segmentation framework for AIAI.

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APPENDIX

APPENDIX A - AI USAGE NOTE

Generative AI tools were used to support ideation and writing. All analytical decisions, data preprocessing, feature engineering, interpretation of results, and final conclusions were made and controlled by the author.

APPENDIX B - CONTRIBUTION STATEMENT

The author was responsible for data preprocessing, feature engineering, clustering analysis, evaluation, interpretation of results, and preparation of the final report. All work was reviewed and finalized by the author prior to submission.

APPENDIX C - RESPONSIBILITY STATEMENT

The author confirms responsibility for the content of this report and certifies that the analyses, results, and interpretations presented are accurate to the best of the author's knowledge and comply with academic integrity standards.